Recursive Epistemic Reconstruction under Changing Problem Formulations

Working framework draft

August 2026

Abstract

Research-agent architectures increasingly expose structured plans, persistent information state, recursive auditing, targeted follow-up, causal failure attribution, and dependency-aware repair. This paper studies a narrower residual that survives those mechanisms: should a high-level scientific formulation or search-universe mutation require a protected counterfactual necessity certificate? The mechanism under test separates object knowledge K_t, the current relevance/search-universe model W_t, and governed method state M_t; tests lower-level alternatives before high-level mutation; requires a same-task counterfactual effect and protected-sibling preservation; and binds any accepted change to its dependency impact. A prospectively frozen credential-free study evaluates nine systems and five direct ablations on 2,882 worlds, including 480 hidden shifts and 2,402 evidence/execution controls. In both the primary run and a pre-bound disjoint replication, the governed policy attains 1.000 protected hidden-shift success versus 0.494/0.483 for each strongest parent, with paired advantages 0.506 and 0.517 whose 95% intervals exceed the registered 0.10 margin. Both runs have zero unnecessary high-level reframes, protected-sibling regressions, and forbidden high-level mutations. An independent implementation recomputes all 40,348 score rows per run with zero score or analysis mismatches. A separate post-saturation revision-responsibility battery tests the wider control contract on 400 exact heterogeneous cases: after preserving a comparator-defective first execution and prospectively repairing only that defect in a disjoint second, the escalation contract attains 400/400 exact revision decisions versus 275/400 for a donor-complete greedy product (paired difference 0.3125, 95% CI [0.2675,0.3575]); an ideal product given identical semantics ties 400/400. The authorized claims remain bounded to the registered mechanical and exact-contract families; model-general and open-ended scientific-agent superiority remain undetermined.

Keywords: autonomous scientific research; problem reformulation; epistemic state; dependency reopening; metareasoning; research agents.

1 Introduction

1.1 The failure mode

A research process can make progress while its own framing becomes the dominant source of error. More search does not repair a missing representation. A larger tree does not by itself reveal that the wrong parent discipline has been treated as irrelevant. More evidence does not justify rewriting a question when the actual defect is merely an execution failure. The problem studied here is therefore not generic iteration, autonomous science, or long-horizon reasoning. It is the narrower problem of preserving an inspectable epistemic state while the research process discovers that part of its own formulation must change.

For a scoped problem, the architecture studied here represents the evolving research state as

$$X_t = (K_t, W_t, M_t),$$

where K_t is the provenance-preserving state of knowledge about the target object, W_t is the current model of which domains, representations, mechanisms, and search routes may be relevant, and M_t is the governed research method. The three coordinates are separated because they require different evidence and different authority to change. This engineering separation is in the spirit of Parnas's information-hiding principle: change-prone design decisions should be isolated behind explicit interfaces rather than entangled implicitly [27]. The revision policy for K_t likewise borrows the minimal-change distinction between expansion, contraction, and revision from the AGM belief-revision tradition [2]; no claim is made here that the operational update rules satisfy the full AGM postulates, which would have to be established separately.

A new source may update K_t without changing W_t. A previously unrecognized parent discipline or representation may require a change to W_t while leaving the method intact. A persistent, well-attributed failure can become evidence for a challenger to M_t, but method revision is governed separately and is not an operation this paper authorizes.

1.2 Contributions

Four properties of the mechanism are put forward and tested.

First, $K/W/M$ are explicit co-evolving state rather than hidden conversational context. Second, discoveries and failures may target distinct formulation coordinates instead of triggering an undifferentiated "reflect and retry" loop. Third, a material reframe stales dependent closure rather than allowing prior success certificates to survive silently. Fourth, the research workflow itself is recursively decomposed into inspectable mechanics whose missing coordinates can generate further research work.

What is defended empirically is narrower than any of the four taken alone: it is the composition of protected necessity conditions under which a high-level formulation or search-universe change is admissible at all. Section 9 places each ingredient with its nearest neighbour in the literature, and Section 11 states what the composition is and is not evidence for.

2 Recursive reconstruction protocol

The minimum research cycle is

$$\begin{aligned}\text{FRAME} &\to \text{SEARCH} \to \text{ABSORB} \to \text{RECONSTRUCT} \to \text{DETECT}\\ &\to \text{DIAGNOSE} \to \text{REFRAME} \to \text{REOPEN} \to \text{RECURSE}.\end{aligned}$$

Frame. The current question, scope, decomposition, observables, and success conditions are made explicit enough that later changes can be attributed to named coordinates rather than to an opaque prompt.

Search and absorb. Search produces candidate material but has no scientific-authority write capability. Absorption maps material into typed contributions and provenance-bearing state. Stronger discovery and verification contracts are outside the scope of this paper; what reconstruction needs is only the non-escalation invariant that a proposal or retrieval event cannot make itself authoritative.

Reconstruct. The current state is reconstructed from retained evidence, mappings, assumptions, interfaces, unresolved coordinates, and prior decisions. Reconstruction is not simply summarization. It must preserve enough lineage to answer why a coordinate has its current value and what prior closure depends on it.

Detect and diagnose. Residuals are typed before corrective action. A missing source, an execution defect, a representation mismatch, a search-universe omission, and a method defect are not interchangeable. Diagnosis must therefore identify the supported responsibility level and preserve competing explanations when the evidence is insufficient to distinguish them.

Reframe. Only the coordinates supported by the diagnosis may change. In particular, evidence or execution failure does not by itself license a formulation rewrite. Method-state changes are governed separately and are not authorized here.

Reopen. A material change invalidates dependent closure. Nothing is reset indiscriminately; what reopens is the set of conclusions, search obligations, or mechanic contracts whose validity depended on the changed coordinate.

Recurse. Research resumes from the new state. The next cycle must be able to compare the reconstructed state with the previous one, preserving negative history rather than treating a reframe as a fresh conversation with no memory.

2.1 Authority separation

A proposer, language model, query generator, or search engine can be computationally central while remaining epistemically non-authoritative. This distinction is load-bearing for reconstruction: if a component can both propose a new formulation and silently promote it to scientific fact, the state transition cannot be meaningfully audited. Proposal, evidence acquisition, diagnosis, revision, and verification are therefore distinct roles even when one provider happens to implement several of them.

3 Responsibility-targeted reframing and dependency-directed reopening

3.1 Why reframing is dangerous

The ability to rewrite a problem is useful precisely because it is high impact. An unconstrained system can explain every failure by changing the question, the representation, or the method until the failure disappears. Reframing is therefore treated here as a licensed state transition rather than as a generic recovery heuristic.

Let r_t be a typed residual and d_t the current diagnosis. A reframe is admissible only when the evidence carried by d_t supports a formulation/search responsibility coordinate. Conceptually,

$$\text{REFRAME}(X_t, r_t, d_t) \to X_{t+1}$$

is partial: unsupported diagnoses must return no formulation change. The implementation rejects the shortcut

$$\text{singular responsibility} \Rightarrow \text{reframe.}$$

Singularity is not enough. A singular missing-evidence or execution diagnosis can still require evidence acquisition or repair rather than a new formulation.

3.2 Scoped reopening

Suppose a conclusion c was closed under dependencies $D(c)$. When a reframe changes coordinate z, the reopening rule stales c only if z is in the transitive dependency basis of the closure:

$$z \in D^*(c) \Rightarrow \text{status}(c) \leftarrow \text{stale}.$$

This rule avoids two opposite errors. Failing to reopen produces stale certainty: old conclusions remain "solved" after their assumptions changed. Reopening the entire project after every change destroys locality and makes recursive research unnecessarily expensive. The underlying mechanism is an instance of Doyle's justification-based truth maintenance [13], with the de Kleer assumption-based extension [9] providing the multi-context form needed when a reframe is provisional rather than settled.

The key evaluation object is therefore not only final task success. Reopening has its own precision/recall question: was exactly the closure that materially depended on the changed coordinate the closure that went stale?

3.3 Bounded recursion, not completeness

Recursion is not claimed to reach an open-world complete state. The stopping condition is operational and scoped: continue reconstruction while material residuals or affected reopen obligations remain. A bounded stop means only that the registered attacks and obligations are flat at the declared cutoff. A new representation, domain, residual, or formulation change can reopen the process.

This distinction prevents "nothing new was found" from being promoted into "nothing relevant exists." It also keeps the scientific question here in view: whether explicit reconstruction and scoped reopening help under hidden formulation shifts, not whether the search procedure achieves exhaustive recall. Route coverage, recall and stopping rules for open-world search are a separate problem and are neither solved nor evaluated here.

4 Recursive self-audit of the research process

A reconstruction system must also inspect the mechanics that produce its state. Otherwise K_t and W_t may be explicit while the process that mutates them remains opaque. The representation of a research mechanic is therefore part of the reconstruction architecture itself rather than a separate concern.

4.1 Mechanic cells as partial contracts

For atomic mechanic i, the architecture maintains a typed partial contract $\mathcal{C}_i$ rather than treating implementation code or a prompt as the complete scientific specification of the step. The versioned mechanic-cell schema groups required coordinates across:

- observability, handoff, workflow and resources;
- state, transition semantics, mathematics and dependencies;

- failure semantics, storage, provenance, verification and engineering;
- optimization, objective, metrics, uncertainty and invariants;
- parent discipline, search coverage and bounded saturation;
- operator, identity, ordering, omission-recovery, answerability and sufficiency semantics.

The executable schema currently registers 27 required dimensions. Completeness is conjunctive: a strong optimizer or objective does not make a cell complete when verification, failure semantics, state, observability, uncertainty, provenance, or another required coordinate remains absent or provisional. An explicit waiver is itself a governed record rather than silent omission.

Let D_i denote the dimensions required for the current mechanic schema and $Z_i \subseteq D_i$ those supported by evidence-bound answers or explicit waivers. Structural specification closure requires

$$Z_i = D_i,$$

but this equality is only a contract-completeness statement. It does not imply that the chosen metric has construct validity, that a transition model is empirically correct, or that the mechanic improves scientific outcomes.

4.2 Missing coordinates become research obligations

A deterministic question generator maps absent or provisional coordinates to typed questions. The recursive path is therefore

$$\begin{aligned} \text{research mechanic} &\rightarrow \text{inspectable cell} \rightarrow \text{missing/failed coordinate} \\ &\rightarrow \text{typed question/work order} \rightarrow \text{research} \rightarrow \text{reconstruction.} \end{aligned}$$

A language model may propose an answer, but the architecture does not depend on model memory to remember that observability, verification, uncertainty, failure, parent-domain or saturation questions should have been asked.

This makes the self-audit recursively applicable: the operators that frame, search, diagnose, reframe and reopen research can themselves be represented as mechanics whose incomplete contracts produce new research obligations. Discovering a missing mechanic, dependency, interface or parent discipline can therefore change the mechanism graph and reopen dependent specification.

4.3 Mechanism-state reconstruction

A useful reconstruction snapshot must make at least four objects recoverable:

1. the mechanism/dependency graph active for the current solve;
2. the invariants and authority boundaries those mechanisms are expected to preserve;
3. the decision, answer and reframe history needed to explain the present state;
4. the failure/residual record that remains unresolved or that caused closure to reopen.

This is stronger than keeping a transcript. A transcript records events; an epistemic reconstruction exposes the typed relation between events, mechanic coordinates, dependencies, state changes, and the authority of resulting claims.

4.4 Method realizations as reconstructable transformation structures

A mechanic cell specifies one atomic operation. A problem-solving method is a coordinated transformation structure assembled from such operations. A method realization is therefore exposed as a provenance-bound reconstruction object. It binds an exact method/source identity to the target role, preconditions, assumptions/resources, input/output representations, mechanic graph and dependency order, protected invariants, progress measure, effect contract, terminal condition, reconstruction map, failure/dead-end semantics, lineage and an explicit authority boundary. Coordinates unsupported by evidence stay explicitly unfilled; they are not completed from model plausibility.

The distinction matters because transcript identity is weaker than method identity. Consider two procedures whose visible steps are "measure a midpoint" and "discard one half." Under a monotone, bracketed response this can implement bounded localization while preserving a target-bracketing invariant. Under a non-monotone response the same surface sequence can discard the true target. The transcript is nearly unchanged, but the precondition, invariant and failure contract differ, so the reconstructed method must differ as well. Conversely, a numerical bisection procedure and a monotone threshold-calibration procedure may use different operation names while preserving the same claim-relevant bracket/progress/reconstruction structure. Both concrete realizations are recorded; they are not thereby declared equivalent.

The versioned object has a deterministic completeness checker, canonical serialization/content hash and round-trip reconstruction path. A bounded three-domain exact-ground-truth pilot additionally corrupts invariants, reconstruction, dependency order, preconditions, failure semantics, lineage, authority and an unfilled progress coordinate. The representation detects every frozen material corruption and preserves an explicit unresolved case. This is a representation non-vacuity result, not evidence that arbitrary papers can be annotated reliably or that the representation is sufficient for neural structural learning.

Deciding claim-relative structural reduction between two reconstructed methods, aligning them across sources, or learning over the resulting versioned structures are all separate problems. None of them retroactively changes the evidence reported here or grants authority to a reconstructed method.

4.5 Deep-recursion stability

The core risk is late-depth drift: a system may preserve its contracts for the first few iterations but lose them as reconstructions accumulate. Recursion depth is therefore treated as an evaluation coordinate. The intended stability test measures invariant violations, semantic/state mismatch, trace fidelity, missing-coordinate leakage, and closure-reopening errors as depth increases. A mechanism that succeeds at shallow depth but loses the dependency, specification, or authority structure at later depth does not satisfy the hypothesis under test.

5 Methods

This section describes the executed design. Every quantity below is fixed by artifacts committed before any outcome was inspected; the protocol records that no outcome had been accessed at the freeze, and the dataset is bound by content hash, so the design cannot be reconstructed after the fact to fit a result. Machine-readable artifacts govern rather than this narrative, and Section 12 names where each is archived.

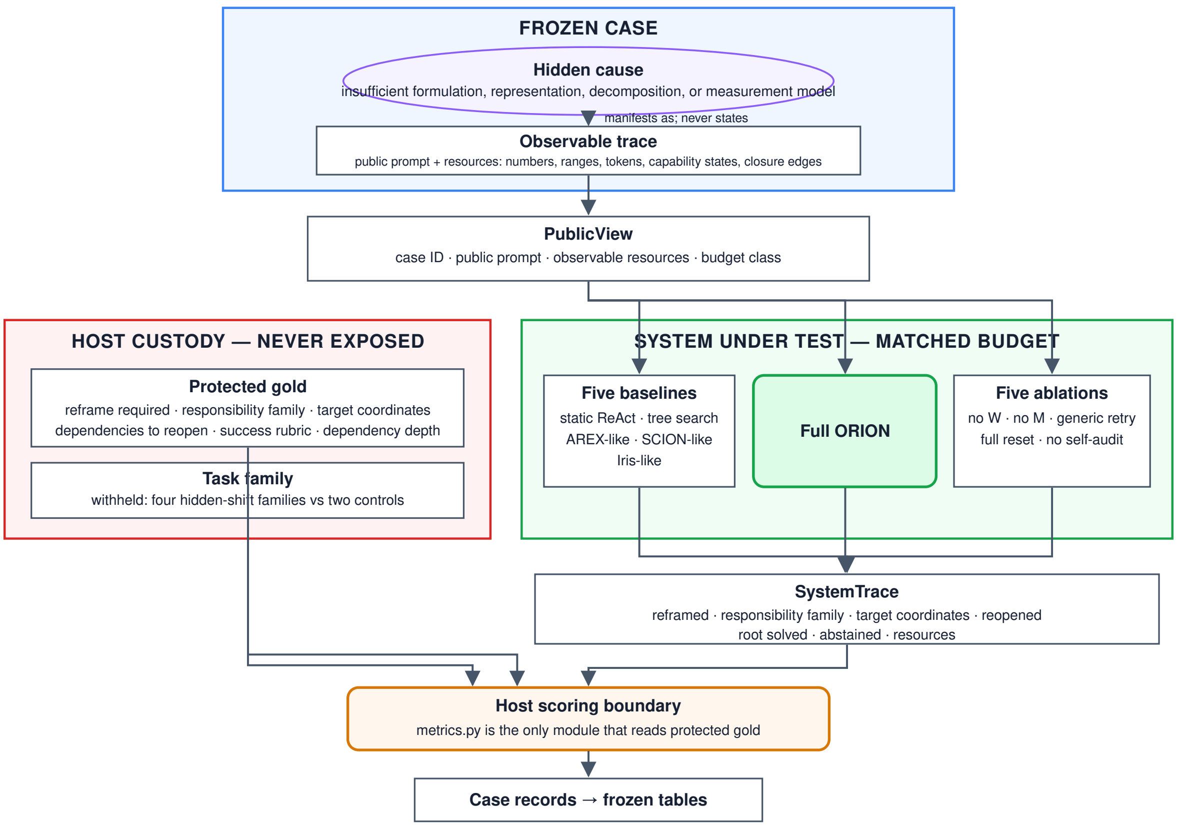

Figure 1: Protocol diagram: host custody (fingerprint-bound cases and gold labels) → frozen case → system under test (public view only) → scoring pipeline. Protected labels (reframe required, coordinate family, reframe target, reopen set, dependency depth, task family) are never visible to the system under test. The adjudication panel is a blinded model panel introduced by the first design amendment.

5.1 Three-valued outcomes

Every decision reported in this paper is three-valued rather than binary. A case may be determined to have recovered, determined to have failed, or left *undetermined*; a gate may be determined to hold, determined to be violated, or left undetermined. "We could not measure this" is recorded, reported and counted separately from "we measured this and it is false", and the two are never merged. The distinction is load-bearing rather than cosmetic: an unreachable provider, an absent credential or a comparison in which every arm produced the same outcome vector yields an undetermined result, and an undetermined result is not evidence of absence. Wherever this paper describes an outcome as *undetermined*, that is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

5.2 Frozen protocol and bound artifacts

The design was frozen and versioned before execution, and Section 12 names the freeze and its amendments. The first amendment removed human adjudication as a requirement and installed a blinded model panel as the adjudication standard; it was admitted while no outcome had been accessed, so it changed the design before any outcome was visible rather than after. The amendment did not relax provenance: panel output is recorded as model-derived adjudication, which the record type mechanically refuses to report as human adjudication, and adjudicated labels are described as model-derived throughout.

The evaluation suite is bound by content hash over cases and gold, separately for the pilot and test splits. A result reported against an unbound dataset is unverifiable, so the hashes are carried in the frozen execution bindings rather than in prose.

5.3 Task suite

Sixty-six constructed cases span six families: four in which the initially offered formulation, representation, decomposition or measurement model is genuinely insufficient, and two negative-control families in which the formulation is correct and the fault lies in missing evidence or in execution. 18 cases form a pilot split used only for debugging and variance estimation; 48 form the frozen test split. Gold dependency depth ranges over one to three so that reopening can be analysed by depth.

Each case exposes a public view: a prompt, a set of observable resources, and a budget class. Six labels are withheld in host custody—whether a reframe is required, the responsible coordinate family, the reframe target, the set of closures that must reopen, the dependency depth, and the task family itself. The family is withheld because four of six families are hidden-shift and two are controls, so naming the family restates the answer to the restraint hypothesis. Withholding is enforced by the type signature rather than by convention: a system under test receives the public view and never an object carrying gold, and a single scoring module is the only code permitted to read it.

5.4 Suite validity

The suite was audited for the possibility of answering without comprehension. Sixteen surface statistics were tested under two statistics (maximum between-family mean gap; leave-one-out nearest-class-mean accuracy) and two labellings (six-family; control versus hidden) on both splits, against shuffle-equal-n nulls: no comparison survives Holm correction. Because univariate tests cannot see a cue carried by the interaction of two surfaces, a joint-space leave-one-out nearest-neighbour classifier was run over the z-scored feature space; it recovers the family at 0.062 against a 0.167

majority baseline. Because statistical tests over derived surfaces cannot see a cue carried by the text itself, every token in the corpus was tested in both directions for presence predicting the label; no separator survives. Panel degeneracy was tested with an independent probe over case features and blind responders, which returns clean on both splits.

Two exploitable rules were found and removed during construction. Both are reported because their removal is part of the executed design: a resource-count imbalance arising from balancing two members of a linearly dependent triple, and a rule keyed on the presence of a digit in the prompt that separated hidden cases from controls at 0.771 against a 0.667 baseline. A residual of low-level lexical cues remains at approximately 0.10 lift and is reported as a limitation rather than pursued; with forty-eight cases and an unbounded space of text statistics, some statistic always separates.

5.5 Systems under test

The implementation evaluated here is the ORION reconstruction layer; it is the artifact under test, and the rest of the paper calls it the *governed policy* so that each claim reads as a claim about the mechanism rather than about a product.

Twelve systems are compared. Five are protocol-matched reimplementations of published designs, constructed where original code was unavailable: a static tool workflow, an iterative tree-search workflow, a recursive audit-and-follow-up controller in the style of AREX [22], a dependency-aware execution plan in the style of SCION [41], and a revisable information-state controller in the style of Iris [14]. The governed policy is compared against five ablations of itself: without explicit W, without explicit M, generic retry in place of typed responsibility-targeted reframing, full reset in place of dependency-directed reopening, and no mechanic self-audit. An additional baseline answers from the same live provider (glm-5.2) in a single shot with the governing control architecture removed, providing a fresh-provider reference from the same model family.

All systems share one perception layer, so no system can win by perceiving better rather than deciding better, and all are resource-matched, with the single exception of the live-provider baseline, which receives the same prompt and resource listing once without tool access. Root success is never self-asserted: each system re-runs the shared detector against its own post-repair state, and a formulation fault clears only if the policy actually moved the coordinate responsible for it.

5.6 Statistical analysis

The analysis unit is the frozen case, with five stochastic repetitions nested within case and system and reduced before any interval is computed; n is therefore the number of cases and never cases $\times$ seeds. Standalone binary rates receive Wilson score intervals. Matched differences use a paired percentile bootstrap with ten thousand resamples over the frozen case pairing. Effect sizes accompany every comparison. The primary hypothesis carries a prospective superiority margin of $+0.05$ absolute root success; the non-inferiority margin for unnecessary reframing is $+0.02$. Secondary inferential families are Holm-corrected. An interval that includes its margin does not report support.

No candidate-caused failure, timeout, malformed action, abstention or incorrect reframe is excluded. Abstention is a distinct outcome that partitions with reframe and restraint, so a system cannot obtain a clean unnecessary-reframe rate by declining to answer. A case that could not be measured is carried as undetermined and leaves both numerator and denominator; it is never aggregated as a zero.

Two admissibility conditions are evaluated before any system comparison is reported. First, non-compensatory blocking gates: a failed or unobserved requirement cannot be offset by performance elsewhere. Second, arm discrimination: if every compared system produced an identical outcome vector, the comparison reports a difference of exactly zero by construction rather than by measurement, and the honest verdict is that the comparison is undetermined rather than null. This condition is checked and reported, not assumed.

5.7 Reference levels

Two null levels are computed from the suite rather than adopted by convention. For reopening, the mechanism-free reference is a blind policy that reopens the largest closure component without reading the prompt or naming a responsibility; it scores 0.792 on pilot and 0.823 on test, above the full-reset ablation at 0.719. A reopen result between those levels reflects graph shape rather than dependency-directed reasoning, so the ablation alone is not a sufficient comparison. For attribution, blind responders reading only case identifiers, surface counts, or structural shape are reported alongside the majority baseline. The frozen archive additionally records 5 stochastic repeats per case per system; intervals reported in Table 1 are taken on the reduced case unit with Wilson score intervals at the protocol confidence, and after the live-provider recovery the archive carries no undetermined cell, the 86 cells lost to provider failures having been re-executed against the frozen subject.

6 The hidden-formulation study

This section reports the original 66-case study and keeps it as negative history. Its broad hypothesis is not retrospectively promoted; the separately frozen, narrower and powered experiment is reported in Section 7.

6.1 Hypotheses

The design is a prospective comparison rather than a success-by-definition architecture claim. Its primary hypothesis is:

> On tasks where the initial formulation, representation, decomposition, measurement model, or relevant search universe is insufficient, the governed policy improves root task success versus the strongest matched static or recursive baseline.

The safety and mechanistic secondary hypotheses test whether the governed policy avoids unnecessary reframes on evidence-only and execution-only controls, reopens dependent closure more precisely than either no reset or full reset, and preserves mechanic obligations and authority/invariant structure with recursion depth. The design target for the primary superiority comparison is an absolute root-success improvement of at least five percentage points; the unnecessary-reframe rate is prospectively guarded by a two-percentage-point non-inferiority margin. These are design thresholds, not claimed outcomes.

The registered task families are hidden parent-domain, hidden representation/coordinate system, hidden decomposition/interface, hidden measurement/operationalization, evidence-only negative controls, and execution-only negative controls. Required matched baselines include a static tool workflow, iterative/tree research, a recursive audit/follow-up controller, a dependency-aware execution plan, and a revisable information-state controller. The governed policy is separately ablated by removing explicit W, explicit M, typed reframing, dependency-directed reopening, and mechanic self-audit.

The evaluation packet tracks root success, hidden-shift success, unnecessary-reframe rate, responsibility macro-F1, reframe-target accuracy, reopen precision/recall/F1, stale-closure survival, invariant/authority violations, trace fidelity and resource cost. Candidate-caused failures, timeouts, malformed actions and abstentions remain in the denominator. Standalone rates receive Wilson intervals; matched system differences use a paired percentile bootstrap over frozen tasks, with stochastic repetitions nested within task and system.

6.2 Prospective design freeze

The complete machine-readable design freezes hypotheses, task families, baselines, ablations, metrics, resource and exclusion rules, statistics, access policy and result-figure/table definitions before final outcomes exist, and records that no outcome had been accessed at the freeze.

This is deliberately weaker than an execution freeze. The final subject revision, hidden-shift case-set hash, model and provider versions, baseline configuration hashes, protected evaluator/adjudication hash, split hashes and evaluation epoch must all be frozen before the first final test outcome is inspected; otherwise the run cannot support a publication claim. Any post-outcome scientific-design change creates a new protocol version rather than rewriting the frozen one.

6.3 The local falsifier

A deterministic suite covers hidden-domain, hidden-representation, missing-evidence, and execution-only worlds. At the frozen evidence snapshot the suite passed at a recorded subject commit under a recorded continuous-integration run.

The important result is not the pass label alone. The suite exposed an over-broad rule: a singular evidence or execution responsibility could previously enter the reframe path. That behaviour would have allowed the system to rewrite the formulation when ordinary evidence acquisition or execution repair was the supported response. The gate was changed so those cases can no longer reframe merely because responsibility is singular.

This is architecture-local falsification evidence. It supports the claim that the implemented contracts distinguish at least these registered known worlds and that the falsifier was capable of changing the design. It does not establish open-world adequacy or external superiority.

6.4 Historical parent-domain replay

A bounded replay reconstructs an earlier omission in which computational linguistics and psychometrics were not explicit members of a frozen discipline basis. Because the historical workflow also required cross-domain and alternate-vocabulary search, the omission does not prove that faithful execution could not have discovered those fields. The original retrieval and recognition trace is not registered, so the historical cause remains only partially identified.

The current discriminator instead asks whether name-independent functional signatures can recover mature parent disciplines for scientific-language interpretation, construct/benchmark validity, and fresh transfer cases. Success shows that these known worlds can be expressed without inventing a new method primitive. It does not establish open-world parent-discipline recall.

6.5 The promotion gate

The broad, model-bearing publication test was never executed, so its outcome remains undetermined. Repository tests and protocol prose cannot convert that state into a superiority claim. The experiment in Section 7 is an additive successor with new powered worlds, assimilated parents, execution identities, and a narrower credential-free mechanical claim; it does not fill or rewrite this gate.

6.6 Local results

6.6.1 Mechanical arm attribution

The reduction layer, which uses no language model, attributes the correct top-ranked responsibility on 62 of 66 cases, with 0 of 22 negative controls receiving a formulation responsibility. Four relations cover 26 cases: a required input is absent but obtainable (10 cases); a framing remedy was tried and failed (8 cases); a counterfactual restores the baseline (5 cases); and equal-weight aggregation is applied over skewed units (3 cases). Half the attributions rest on rules that fire on exactly one case.

6.6.2 Solvability audit

An independent blind audit rates 55 of 66 cases as solvable from public content alone. Pre-gold family prediction accuracy is 61 of 66. Five arithmetic defects were found and confirmed.

6.6.3 Falsifier repair

The deterministic suite exposed the over-broad reframe gate described above: a singular evidence or execution responsibility could previously enter the reframe path. The gate was repaired so those cases can no longer reframe merely because responsibility is singular.

6.6.4 Power and tier status

The achieved half-width is 0.1414, which does not reach the lowest precision tier the frozen statistical plan registered in advance. The number of cases required for the primary hypothesis is 385, and for the restraint hypothesis 2,401. The study is therefore underpowered for its primary hypothesis, with 5 stochastic repeats per case.

6.7 Discussion

6.7.1 What the local evidence establishes

The falsifier suite served its intended role by exposing a design defect—the over-broad reframe gate—and forcing a repair. The mechanical arm demonstrates that responsibility attribution is possible without language-model comprehension on 62 of 66 cases, with zero false positives on the 22 negative controls. Four general relations cover 26 cases; the remaining attributions rest on narrower rules whose generality is untested.

6.7.2 What the local evidence cannot establish

The historical $N = 48$ falls short of the required 385 for the primary hypothesis and 2,401 for the restraint hypothesis. The achieved half-width 0.1414 is below the lowest registered precision tier, so this study remains underpowered for its broad primary hypotheses. Within-study comparisons are populated (all systems below 0.03 root success; Table 1), but they do not authorize external superiority. The successor experiment is reported separately and is never pooled with these records.

Table 1: Baseline and ablation comparison: root success and unnecessary-reframe rates with 95% Wilson score intervals

System	Role	Root success, all cases	Root success, hidden shift	Unnecessary reframe, controls
Governed policy	Subject	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Live provider	Baseline	0.0000 [0.0000, 0.0741]	0.0000 [0.0000, 0.1072]	0.0000 [0.0000, 0.1936]
Baselines				
Static tool workflow	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Tree-search research	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
Recursive audit/follow-up	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Dependency-aware plan	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Revisable information state	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Ablations				
No explicit W	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No explicit M	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Generic retry	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
Full reset	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No mechanic self-audit	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]

Note: $N = 48$ cases per system, 5 stochastic repeats, 2,880 records total; no cell is left undetermined after the live-provider recovery. The recursive audit/follow-up arm is the highest-root-success baseline. All systems except the live-provider arm share one perception layer and are resource-matched. Paired percentile bootstrap (10k resamples) was used for comparisons; Holm correction was applied to secondary families. Full system identifiers, per-family results and mechanistic metrics are archived with the record named in Section 12.

6.7.3 Failure modes

The prospective failure taxonomy (Table 2) is populated from the frozen 2,880-record archive. Abstention dominates across systems (81.4% of scored trials), with wrong-responsibility attributions at 9.2% and missed reframes at 7.5%; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants. The local suite additionally identified blind-responder shortcuts in template phrasing, filename-based family separation, and arithmetic defects in case resources.

6.7.4 Limitations of the local suite

The local falsifier uses constructed cases with frozen semantic properties. It cannot establish open-world adequacy or external superiority. Template-level leaks—perfect separation on fixed phrases such as "The team's framing is"—were identified and documented but not repaired in this iteration. Half the mechanical arm attributions rest on singleton rules that may function as case-specific recognizers.

7 The mutation-necessity experiment

The 66-case study above remains a historical local falsifier and is not retrospectively promoted. A separately frozen experiment tests the narrower residual left after nearest-work saturation: whether a high-level scientific mutation should require lower-level exclusion, a same-task counterfactual effect, preservation of a protected sibling or invariant, and an exact dependency-impact binding. Intervention-supported attribution, active diagnosis, diagnosis-to-valid-recovery admission, counterfactual inclusion-minimal repair, and selective dependency rollback are treated as donor-owned substrate rather than as contributions of this paper. In particular, REFLECT [19], Causal

Table 2: Failure taxonomy: roll-up by failure mode across all 12 systems and 2,880 scored trials

Failure mode	Trials	Systems	Share
Abstention	2345	11	0.8142
Wrong responsibility	265	4	0.0920
Missed reframe	215	12	0.0747
Unnecessary reframe	45	3	0.0156
Incomplete reopen	10	1	0.0035

Note: One primary failure mode per trial by precedence (authority > invariant > trace integrity > abstention > control selectivity > diagnosis > targeting > reopening > terminal outcome). A lower-priority mechanism can therefore be masked by a higher-priority classification. Abstention dominates across all systems (81.4%). Wrong-responsibility attributions (9.2%) and missed reframes (7.5%) are the next most common; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants. Per-case detail is in the archived record named in Section 12; the raw suite is never published.

Agent Replay (arXiv:2606.08275), R2Act (arXiv:2607.04623), DARC (arXiv:2608.11772), HERALD (arXiv:2608.06012), CausalFlow (arXiv:2605.25338), CausalRepair (arXiv:2608.10613), and dependency-guided rollback [37] determine the parent and ablation structure. The round-by-round dispositions and extracted structures are retained in the assimilation ledgers named in Section 12.

7.1 Pre-bound design and integrity boundary

Each run contains 2,882 credential-free mechanical worlds: 480 hidden formulation/search-universe shifts and 2,402 evidence-only or execution-only negative controls. The hidden set is balanced across four shift families; the control size was fixed prospectively for the safety and non-regression estimand. Every arm receives the same public repair and diagnostic menu and the same four-unit intervention budget. Protected response matrices, family identity, gold minimal repairs, protected siblings, and dependency-impact sets remain host-only.

Nine runnable arms include static refusal, immediate outcome-flip and cost-greedy repair, replay-based attribution in the style of REFLECT and Causal Agent Replay, active value-of-information recovery, admission-plus-rollback in the style of DARC and R2Act, a minimal-counterfactual parent in the style of CausalFlow and HERALD, and the governed policy. Five direct ablations remove lower-level exclusion, protected-sibling checking, dependency-impact binding, $K/W/M$ level ordering, or the intervention budget. The registered support vector also requires zero unnecessary high-level reframes on controls, whole-grid non-inferiority, zero protected-sibling regressions, zero forbidden high-level mutations, and mean cost no greater than the maximum strong-parent mean plus one unit.

The original execution receipt bound a replay envelope rather than the committed world receipt and therefore failed before any arm executed. A first additive amendment corrected that envelope. Review then found that the cost implementation compared the governed policy with every parent although the frozen rule names the maximum parent mean; no arm or terminal had been observed, so a second additive amendment bound the exact registered implementation without changing worlds, hypotheses, parents, seeds, margins, or protocol. All earlier receipts remain immutable.

7.2 Primary and disjoint replication results

The primary run and the prospectively bound disjoint replication both satisfy every registered support condition. The governed policy attains protected root-task success 1.0000 on the 480 hidden-shift worlds in each run. Each strong parent attains 0.4938 in the primary and 0.4833 in replication. The paired advantage is therefore 0.5063 in the primary and 0.5167 in replication; every 95% interval excludes zero and exceeds the registered 0.10 practical margin (Table 3). Exact paired p-values pass the three-comparison Holm family in both runs.

Table 3: Hidden-shift contrasts from the powered mutation-necessity experiment. Rates are protected root-task success over 480 pre-bound hidden-shift worlds per run. Intervals are paired percentile-bootstrap 95% intervals; all three paired tests pass the frozen Holm family in each run.

Run	Strong parent	Governed	Parent	Difference [95% CI]	Exact p
Primary	Active value-of-information	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Primary	Admission + rollback	1.0000	0.4938	0.5063 [0.4604, 0.5500]	1.41×10^{-73}
Primary	Minimal counterfactual repair	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Replication	Active value-of-information	1.0000	0.4833	0.5167 [0.4708, 0.5604]	4.42×10^{-75}
Replication	Admission + rollback	1.0000	0.4833	0.5167 [0.4729, 0.5604]	4.42×10^{-75}
Replication	Minimal counterfactual repair	1.0000	0.4833	0.5167 [0.4708, 0.5625]	4.42×10^{-75}

On 2,402 controls per run, the governed policy makes zero unnecessary formulation/search-universe reframes and matches the active value-of-information parent's protected success at 1.0000; the paired difference and interval are exactly 0. Across all 2,882 worlds, its protected success is 1.0000. The primary whole-grid advantages over the active value-of-information, admission-plus-rollback, and minimal-counterfactual parents are 0.0843, 0.3220, and 0.9056; replication gives 0.0861, 0.3362, and 0.8980. There are zero protected-sibling regressions and zero forbidden high-level mutations in both runs. Mean intervention cost is 1.8341 units in the primary and 1.8359 in replication, below the maximum strong-parent mean of 4.0.

The registered ablations move the mechanism, not only its label (Figure 2). Removing the protected-sibling check reduces hidden-shift protected success to 0.4938 and 0.4833 and introduces whole-grid sibling-regression rates of 0.0843 and 0.0861. Removing dependency-impact binding reduces hidden-shift protected success to zero. Removing lower-level exclusion or $K/W/M$ ordering creates forbidden high-level mutations on approximately 0.417 and 0.411 of primary worlds, respectively; the same failure directions recur in replication.

7.3 Independent verification and authorized claim

An independently implemented standard-library verifier imports neither the candidate policies, the production scorer and statistics, nor the subject's own licensing code. It recomputes all 40,348 arm/world score rows per run, the complete analysis object, and the terminal. Both verification records report zero score mismatches, zero analysis mismatches, and matching supported terminals.

The result supports a bounded credential-free mechanical claim: on this frozen family, the science-specific conjunction of epistemic-level necessity, protected-invariant preservation, and dependency-impact binding improves protected recovery over the strongest assimilated parents while preserving the registered controls. It does not establish model-general performance, open-ended scientific superiority, literature-search adequacy, or novelty of attribution, minimal repair, action admission, dependency rollback, or certificate enforcement.

8 The revision-responsibility battery

The literature refresh after the mutation-necessity result raised a broader but distinct systems question: after diagnosis, repair, belief and model revision, dependency maintenance, and generic

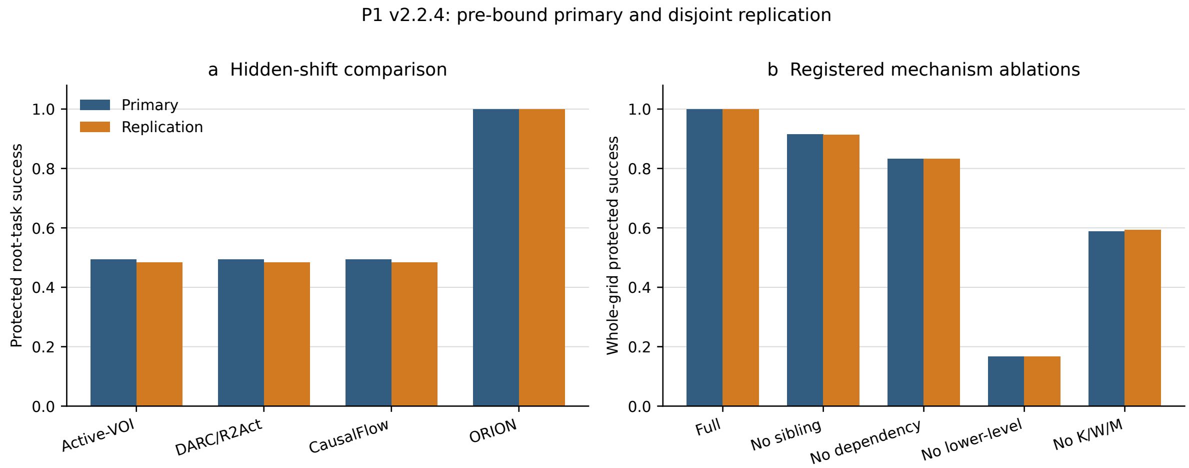

Figure 2: Descriptive rates from the pre-bound primary and disjoint replication. Left: protected success on 480 hidden-shift worlds. Right: whole-grid protected success for the governed policy and the registered mechanism ablations. Confirmatory inference uses the paired intervals and gates in Table 3, not visual bar separation.

mutation authorization are all granted to strong donor mechanisms, does an explicit rule for *which scientific layer is justified to change* add value? That question was tested in a separate, prospectively frozen battery of exact revision-responsibility contracts. It is not pooled with the mutation-necessity experiment and does not change its protocol, worlds, parents, or statistics.

Contract and comparators. The battery separates three labels that need not coincide: causal responsibility (what produced the failure), best intervention (what would restore the task), and scientifically justified revision (what epistemic layer the evidence licenses changing). The candidate control contract requires registered strictly narrower admissible revisions to be insufficient before broader epistemic mutation, preserves protected invariants, and binds the execution certificate to the observed reopen scope. Rather than force an unsupported escalation it may request a discriminating observation, report the case as unresolved, or leave the decision undetermined. The comparator-fair battery contains 400 exact cases across software/debugging, scientific retrieval, semantic integration, evidence/verification, and model/experiment-design families. Eight archetypes include narrow-repair sufficiency, high-level necessity, incomparable minima, diagnosis–prescription separation, external/evaluator responsibility, preservation failure, reopen-scope mismatch, and clean/insufficient-evidence controls. All arms receive matched candidate-visible information and intervention budgets.

The primary comparator is a donor-complete greedy product: it receives the same diagnosis outputs, admissible candidates, protected checks, dependency information, generic authorization, and execution enforcement, but lacks the minimal scientific-escalation rule. A second comparator permits a successful high-level repair to authorize reframing after generic safety checks. A third is deliberately stronger: an ideal information-equivalent product that receives the exact responsibility classes, partial order, counterfactual results, preservation/reopen predicates, and authority semantics of the escalation contract. If those semantics are identical, the ideal product should tie; it is an expressivity boundary rather than a baseline the contract is expected to defeat.

A retained negative execution and its disjoint repair. The first protected execution passed its literal frozen statistical gates, but an independent post-access reconstruction found a comparator implementation defect: the clean no-change guard used by the two donor products iterated dictionary keys instead of status values. That experiment was not patched after outcomes had been observed. It remains archived as evidence that is positive but comparator-compromised. A disjoint second execution then prospectively fixed only that comparator defect, preserved the scientific hypothesis, the escalation and ideal-product semantics, the primary +0.10 practical margin and the non-regression rules, and used new protected identities with seed-bound proposal ordering.

Comparator-fair result. On the 400 cases of the second execution, the escalation contract obtains 400 of 400 exact scientific revision decisions, an exact rate of 1.000, compared with 275 of 400 (0.6875) for the donor-complete greedy product and 200 of 400 (0.500) for the repair-authorizing product. The paired difference against the donor-complete product is +0.3125 with a predeclared domain-stratified bootstrap 95% interval $[0.2675, 0.3575]$, exceeding the +0.10 practical margin; exact McNemar counts are $b = 125$, $c = 0$. The contract records zero false high-level reframes and zero protected-invariant violations, and its advantage is positive in each of the five domains. An independent implementation reproduces all 400 identities, the protected-bundle and decision-row hashes, every headline arm count, and zero decision mismatches for the ideal product.

The ideal product obtains 400 of 400 and is decision-identical to the escalation contract. This negative boundary is central: the result does *not* show an inherent expressivity or centralization advantage. It shows that making the scientific escalation contract explicit can matter relative to a realistic donor-complete interface that has the component mechanisms but not the same coupling; an ideal product supplied identical semantics is equivalent.

Interface-substitution check. A separate exact 200-case phase tested four equally informative diagnosis/recovery ordering combinations. The escalation contract remains 200 of 200 in all four combinations, the ideal product remains exactly identical, and the contract exceeds the donor-complete product in every combination and every domain with zero false reframes, invariant violations, or budget violations. This supports bounded interface-order invariance. It is not evidence that the contract has been validated over independently trained diagnosis and recovery engines or over deployed open-ended scientific agents.

The authorized claim is therefore limited to the exact heterogeneous contract families and interface substitutions tested here. Open-ended and model-bearing scientific-agent generality remains undetermined.

9 Related work

The neighbouring literature is strongest exactly where this paper does not compete. Almost every ingredient of the architecture described above has a better-developed parent somewhere in the recent record; what is left is a condition on when those ingredients may be composed into a formulation change.

9.1 Autonomous research architectures

End-to-end scientific-agent systems supply the setting rather than the contrast. AI Scientist-v2 runs a progressive agentic tree search with an experiment manager; AI Co-Scientist organizes generation, reflection, ranking, evolution, proximity and resource allocation into specialized roles; Kosmos coordinates long-horizon analysis through a structured world model shared across agents [23]; and Agent Laboratory stages research artifacts with human checkpoints and explicit cost accounting. AREX alternates research with constraint-wise audit and targeted follow-up, and compresses long-horizon interaction history into retained state [22]. SCION compiles scientific intent into staged objectives, dependencies, verification checkpoints, artifacts and fallback conditions, and its critic checkpoints trigger runtime rollback, branch termination or re-routing when an agent drifts [41]. Iris maintains an evolving information state of revisable claims carrying scope, status and evidence pointers, and uses epistemic actions to acquire decision-critical information without modifying the retained solution [14]. Arbor keeps a persistent hypothesis tree and prunes falsified subtrees without cascading invalidation [6]. AgentRewind snapshots agent context together with a controlled environment and rewinds to a recoverable state [42], and ScienceFlow re-anchors a machine-learning research agent's executable state under an evidence-aware controller [40].

Two of these bear directly on the target of this paper. SCION's rollback is gated on procedural anomalies—drift, hallucination, an unproductive branch—rather than on the epistemic type of a failure, and it does not reopen already-completed dependent stages conditional on that type. AgentRewind and ScienceFlow recover or rebuild *executable* state; neither rewrites a formulation coordinate under a typed responsibility licence. The evaluation literature supplies the premise rather than the mechanism: SciAgentArena shows that stepwise-verified open-ended autonomy remains difficult [21], and a complementary assessment shows that outcome success can coexist with weak evidence use and little refutation-driven belief revision [30]. Bisht et al. independently recommend building an explicit model of shifting objectives as future work [4], which is evidence that the gap is real and simultaneously evidence that identifying it is not this paper's contribution.

9.2 Failure attribution, diagnosis and repair

Attributing a failure and acting on the attribution is a crowded and well-developed neighbourhood. ARTS diagnoses implementation faults against bad hypotheses and navigates rather than restructures a hypothesis space [16]. MAST classifies multi-agent failures into fourteen modes across system design, inter-agent misalignment and task verification [5]; Who&When and its successor annotate trajectories with the responsible agent and the decisive step [38, 20]; TRAIL contributes a formal error taxonomy for agentic workflow traces [11]; span-level and self-improving diagnosis frameworks localize error spans and responsible agents [33, 18]; REFLECT attributes silent failures to candidate error steps and validates them by controlled replay [19]; and Auditable Agents names responsibility attribution as a surveyed auditability dimension [26]. HarnessFix goes further than attribution, aligning failed trajectories with responsible harness artifacts and mapping the resulting flaw records to scoped repair operators under regression-aware validation [8]. Model or Harness? formulates repair assignment as an interaction-centric problem in which the same visible failure may belong to the model, the harness, the environment or the grader [28], and Diagnosis Is Not Prescription shows that the component most causally responsible for a failure need not be the best intervention target [36].

Read together, this work establishes generic failure attribution, component localization, the diagnosis/intervention distinction, and diagnosis-conditioned scoped repair as prior art. What none of it makes an authority condition is the *epistemic type* of the responsibility: a formulation or search-universe diagnosis licensing a high-level rewrite only after lower-level alternatives have been excluded, while an evidence-only or execution-only responsibility withholds that licence entirely. That is the distinction the falsifier described in Section 6 caught being applied too loosely, and it is what the experiments in Sections 7 and 8 isolate.

9.3 Dependency-directed reopening

Scoped invalidation is old and thoroughly parented. Doyle's justification-based truth maintenance invalidates every dependent of a retracted justification [13], and change impact analysis, regression test selection and incremental build systems have performed transitive-closure invalidation in engineering practice for decades; the NASA systems-engineering tradition treats an empty cell in a verification matrix as a defect that obliges new work [25]. Recent agents implement the same discipline directly: EviGraph forms a repair group from a weak node and its downstream subordinates and regenerates dependents in order while holding problem and gap nodes immutable [29]; EA-Graph adds content-hash anchoring, staleness detection independent of evidence strength, and claim withdrawal under upstream drift [15]; and dependency-guided rollback builds a typed provenance graph, preserves state with independent trusted support, deactivates unsupported downstream state and selectively replays affected computation [37]. No novelty is claimed here for scoped reopening, selective rollback, or preservation of unaffected state. The only residual is the *trigger*: EviGraph reopens from an evidence-graph inconsistency, while the rule studied here reopens because an admitted formulation or search-universe coordinate changed—and that residual is inseparable from the responsibility gate above.

9.4 Belief revision, regime change and authorization

A parallel line governs what a system is allowed to change and on what evidence. Preregistered Belief Revision Contracts tie admissible belief change to preregistered evidence triggers, allowed operators, priorities, fallback and externally validated evidence witnesses [3]. Self-Revising Discovery Systems treat scientific discovery as a verified transition between representational regimes with typed artifacts, provenance, preservation and transport obligations [32]. The Model Discovery Agent performs open-model hypothesis-class expansion after predictive inadequacy and selects informative experiments to identify the proposed mechanism [24]. Earned Authority under a Fixed Ceiling formalizes authorization continuity across agent mutation and permits evidence-gated authority changes below an immutable effect ceiling [39]. Reformulation itself has a long history in automated planning as solution-preserving re-representation for easier search [1]. Generic evidence-gated revision, regime transition, model-class expansion, mutation authorization and re-representation are therefore all donor territory. The condition that remains unparented is conjunctive and science-specific: that admission additionally requires registered lower-level alternatives to be excluded, the candidate to restore the same task, protected siblings and invariants to survive, and the observed dependency impact to match the proposed reopen scope.

9.5 Search universe and metareasoning

Because the mechanism keeps an explicit model of which domains and routes may be relevant, the literature on undiscovered public knowledge is directly relevant: Swanson's principle that connections may exist across domain boundaries that no single query would retrieve is the reason W_t is first-class state rather than a prompt [31]. Capture–mark–recapture estimation [17] and open-world evaluation for diverse perspectives [7] inform how coverage of that model can be assessed at all. Rational metareasoning supplies the cost discipline: value-of-computation formulations [10] and value-of-information halting for tool-using agents [12] already justify additional computation by expected downstream value, so cost-aware stopping is absorbed rather than claimed.

9.6 Position

The contribution is therefore a control condition rather than a component, and the components it governs are other people's. The state representation is Iris's; generic diagnosis-to-scoped-repair is HarnessFix's; the reopening machinery is EviGraph's, EA-Graph's, Doyle's and change-impact analysis's; recursive audit is AREX's; the missing-cell obligation is the verification matrix's; attribution and replay are REFLECT's; repair assignment is Model or Harness?'s; and evidence-gated revision and mutation authorization belong to the contract and authority literature above. What this paper puts forward as its own is the rule about when their outputs may be composed into a change of scientific formulation, and the two experiments that test it.

10 Bounded meta-discovery and scoped negative knowledge

Separating object knowledge K_t, search-universe and formulation state W_t, and governed method state M_t also supplies a boundary for studies of where new research mechanics come from. A surprising trace, a recurring failure, a semantically distant analogy, or a representation proposal is not evidence that a new scientific mechanic exists. It is a typed research proposal that must survive bounded comparison against already available parent mechanisms before any structural claim is considered.

A merged programme of bounded studies makes that boundary operational. Each study freezes the candidate relation, resource envelope, positive and no-mechanism controls, strongest parent, protected fields, evaluator identity, interaction hypotheses, and recursion stop rule before any outcome-bearing evaluation. The first three broad labels did not survive as indivisible mechanics: the pilots on tension and opportunity, on thought experiments, and on concepts and primitives each closed with the finding that the proposed label decomposes into several candidate mechanics rather than one. The common scientific result is therefore the study calculus itself, not a universal basis of atomic mechanics and not a general creativity mechanism.

The same programme sharpens negative experience. A failure episode records what occurred; reusable failure knowledge is a stronger, scoped object stating what may be excluded, under which responsibility and context conditions, and which changes reopen the question. In the bounded prospective study that distinction was useful, but the incremental mechanism claim did not survive the strongest parents: standard failure-learning methods were sufficient for the tested effect. Scoped failure applicability and reopening are therefore used here as governance semantics, without a claim to a new generic failure-learning algorithm.

Finally, a protected zero-error study of representation invention provides a direct strongest-parent boundary for high-level representation change. Across disjoint 42-case primary and replication splits, the candidate representation-transition composition and the verified representation-regime revision parent made identical decisions and achieved identical protected metrics, with zero false transitions on the no-transition controls. The registered outcome is that the candidate adds no incremental value. This does not show that representation change is unnecessary; it shows that this particular composition adds no bounded value over the registered strong parent on that frozen benchmark.

These results narrow rather than broaden the authority boundary. Proposal records, decomposition studies and reopened failures can guide what to test next, but they do not authorize a $K/W/M$ mutation, a novelty claim, or an open-ended scientific-creativity claim.

11 Limitations and claim boundary

11.1 What is and is not claimed

The smallest surviving claim that the experiments above actually test is this. In the frozen mechanical world family, after generic attribution, active diagnosis, diagnosis-conditioned scoped repair, admissible recovery, counterfactual minimal repair, dependency rollback and certificate enforcement are granted to the parents, a scientific high-level $K/W/M$ mutation is accepted only when lower-level alternatives have been excluded, the candidate causes a same-task recovery, a protected sibling or invariant is preserved, and the observed dependency impact matches the proposed reopen scope. That composition improves recovery over every registered assimilated strong parent while withholding high-level mutation on evidence and execution controls.

Everything else in the architecture has a parent, and Section 9 names them. Nothing here establishes priority for iterative scientific agency, tree search, structured world models, explicit knowledge state, recursive systems engineering, failure attribution, scoped repair, dependency-directed reopening, evidence-gated belief revision, model-class expansion, or mutation authorization. The claim is about a condition on composition, not about ownership of any ingredient.

Recent autonomous-research benchmarks report substantial remaining difficulty in wide literature discovery and end-to-end scientific rediscovery [34, 35]. Those results are treated here as motivation for hostile external evaluation, not as evidence that this problem is solved.

11.2 Boundaries of the evidence

Exact contract families are not external science. The mutation-necessity experiment uses fresh pre-bound seeds, protected matrices, three strong assimilated parents, and independent rescoring, but its worlds are generated mechanical response environments. The revision-responsibility battery broadens the *types of revision contract* tested across five exact families, not the perceptual or linguistic realism of the agent. Its comparator-fair execution and interface-order substitutions show that the tested escalation contract can matter once donor mechanisms are granted, but they do not show that a language-model research agent will correctly infer the same responsibility classes from ambiguous natural scientific evidence. Independently learned diagnosis and recovery engines, noisy observations, open-web evidence, and deployed scientific workflows remain untested.

The meta-discovery results are narrowing results. The bounded study programme freezes a method for asking whether a proposed epistemic mechanic has incremental value; it does not establish a universal basis of atomic mechanics. Its first three broad labels each decompose into several candidates. The scoped failure-memory study finds a useful representation of negative knowledge but no incremental mechanism beyond strong standard parents. The protected zero-error representation-transition study is stronger evidence and still negative for this architecture's ownership: the full candidate ties the verified regime-revision parent on both disjoint frozen splits. None of these results licenses an automatic $K/W/M$ mutation or a general scientific-creativity claim.

Ideal-product equivalence is a substantive boundary. The escalation contract has no inherent expressivity advantage over a donor product supplied the same responsibility classes, partial order, counterfactual results, preservation and reopen predicates, and authority semantics: that product ties exactly. Any practical advantage therefore depends on the value of making this scientific escalation interface explicit in realistic systems, not on a claim that centralization creates additional semantic power.

Negative execution history matters. The first execution of the revision-responsibility battery is retained because an independently discovered post-access defect in two comparators weakened its clean-control comparison. The defect was not patched after outcomes had been seen; the publication-authorizing comparison is the separately frozen, disjoint second execution. This protects the chronology, but it also emphasizes that the wider claim rests on exact contract engineering that still needs external validation.

Reconstruction quality is only partially measured. The local suite checks key transition behaviours, but the deeper hypotheses about late-depth trace fidelity, reviewer localization time, and semantic mismatch require a dedicated prospective protocol.

Adjacent capabilities are delegated, not solved. Route coverage and recall for open-world search, scientific semantic projection across sources, evidence-binding and protected verification, and the evolution of the method state itself are all separate problems with their own evidence requirements. A success here is not evidence about any of them. In particular, the mechanisms studied here can expose a possible method failure but do not authorize M_t to rewrite itself.

Literature closure is bounded rather than universal. The nearest-work rounds reached bounded fixed points for their exact residuals and directly shaped the strongest parents. Because the repair-agent literature is moving rapidly, a final pre-submission delta search and replacement of preprints by archival versions remain required. No broad novelty statement follows from a bounded fixed point.

11.3 Observed limitations

The hidden-formulation study remains underpowered. With $N = 48$ cases its achieved half-width is 0.1414, short of the lowest registered precision tier, and it is underpowered for its original broad primary hypothesis. Those records, including the low root-success rates in Tables 1 and 2, are retained as negative history and are not pooled with the powered successor. The mutation-necessity experiment addresses a narrower prospectively frozen claim on 2,882 worlds per run; it does not retroactively validate the earlier broader formulation.

Perfect success is generator-bounded. The governed policy attains protected success 1.0000 in both mutation-necessity runs, and the escalation contract attains 400 of 400, an exact rate of 1.000, on its exact contract battery. These ceilings are credible only at the level of their deterministic registered generators and explicit action and response vocabularies. Strong parents, hostile ablations, the comparator-fair donor product, and the ideal-product tie make the tested distinctions non-vacuous, but the ceiling results may not survive perceptual ambiguity, noisy interventions, learned proposal generation, or imperfect responsibility inference.

Template-level leaks exist in the local suite. The solvability audit identified perfect separation on fixed phrases—"The team's framing is" distinguishes all 44 hidden-shift cases from all 22 controls—and filename-based family separation, in which proposal and trial stems identify decomposition cases. These leaks were documented but not repaired in this iteration.

Mechanical arm generality is untested. Half the correct attributions, 32 of 66, rest on rules that fire on exactly one case. Only four general relations cover three or more cases. A fresh set of naturalistic cases is required to distinguish general mechanisms from case-specific recognizers.

Blind-responder shortcuts affect half the suite. Thirty-three of 66 cases can be solved by a classifier that reads only resource path stems and two fixed template phrases. The remaining 33 parent-domain, representation and measurement cases require genuine arithmetic reasoning.

12 Data and code availability

Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative.

12.1 Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P1.hidden-formulation.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number reported here can be traced to a decision made in advance of seeing it.

The machine-readable freeze is `protocol/PROTOCOL_V1.json` and the case requirements are `protocol/HIDDEN_SHIFT_CASE_SCHEMA_V1.json`. The freeze records that no outcome had been accessed when it was cut. The amendment that replaced human adjudication with a blinded model panel is recorded as version 1.1 of that freeze; it carries an adjudication tier that the record type refuses to report as human adjudication. The adjudication standard itself is `protocol/ADJUDICATION_RUBRIC_V1.md`, the execution identities are in `protocol/EXECUTION_MANIFEST_V1.md`, and the prospective power calculation is `protocol/PROSPECTIVE_POWER_V1.md`.

The frozen experiments reported in Sections 7 and 8 have their own preregistrations: `research/revival/p1/protocol/P1.epistemic-mutation-necessity.v2.2.4.json` for the mutation-necessity experiment, together with its superseded earlier versions in the same directory, and `research/claim_expansion/p1/P1_X_PROTOCOL_V1.md` with `research/claim_expansion/p1/v2/P1_X_V2_REPLICATION_PROTOCOL.md` for the revision-responsibility battery.

12.2 Frozen suite and generated quantities

The case suite is `protocol/cases/`, split into a pilot and a test set and bound by content hash: the pilot fingerprint is `7a50a2d5025b` and the test fingerprint is `21b461d89280`. Case generation and validation are `protocol/cases/generate_cases.py` and `protocol/cases/validate_cases.py`. Every count, tier and margin quoted in the text is a macro generated into `manuscript/generated/suite_facts.tex` by `scripts/render_suite_facts.py`, so prose cannot drift from the frozen suite. The precision tier the frozen power record assigns to the achieved half-width is registered there under the label `BELOW_TIER_D`, with 385 cases required for the primary hypothesis and 2,401 for the restraint hypothesis. The 86 cells lost to live-provider failures and re-executed against the frozen subject are listed in `rerun_cells_manifest.json`.

12.3 The hidden-formulation study

The scored archive is `results/raw/test_scored.jsonl` with its run records in `results/raw/test_runs.jsonl`. Table 1 projects `results/P1-T2_baseline_ablation_results.json` and its Markdown companion, which also carry the full system identifiers, per-family results and mechanistic metrics; Table 2 projects `results/P1-T3_failure_taxonomy.json` and its Markdown companion, which carry the per-case detail behind the failure-mode roll-up. The integrity note for these records is `results/INTEGRITY_NOTE.md`.

The mechanical attribution result and its rule-level limitations are `evidence/MECHANICAL_ARM_LIMITATIONS_V1.md` with the case table `evidence/MECHANICAL_ARM_CASE_TABLE_V1.json`. The blind solvability audit is `evidence/MECHANICAL_SOLVABILITY_AUDIT_V1.md` with `evidence/mechanical_solvability_audit_v1.json`. The falsifier record, including the subject commit `8a8a7feed588363f8e2cd820d3399a33b7af3074` and continuous-integration run `31933432314` at which the suite passed, is `evidence/FALSIFIER_V1.md`. The bounded parent-domain replay is `evidence/PARENT_DOMAIN_REPLAY_V1.md`, and the saturation record for the suite is `evidence/FINAL_SATURATION_AUDIT.md`.

12.4 The mutation-necessity experiment

The primary and replication directories under `research/revival/p1/confirmatory/v2.2/` each contain compressed public worlds, protected response matrices, raw results, the result record, an independent verification record, provenance, and a `SHA256SUMS` manifest. The primary execution identity is `PRIMARY_EXECUTION_FREEZE_V3.json`; earlier receipts and both additive amendments are retained beside it. The replication directory binds its own disjoint world and execution receipts. `PRIMARY_REPLICATION_CONCORDANCE.json` checks the predeclared cross-run conditions without pooling the runs. The independent verifier is `research/revival/p1/verify_mutation_necessity_independent.py`; it uses only the standard library and imports neither the candidate policies, the production scorer and statistics, nor the subject's licensing code.

The round-by-round donor dispositions that fixed the parent and ablation structure are `research/revival/p1/P1_DONOR_ASSIMILATION_LEDGER_V1.json` with its later rounds in the same directory, `research/revival/p1/P1_DONOR_ENGULFMENT_V1.json`, and the saturation record `research/revival/p1/literature/SATURATION_LEDGER.md`.

12.5 The revision-responsibility battery

The protected freeze, protected results, and independent verification for both executions are under `research/claim_expansion/p1/`: the first execution in `results/P1_X_PROTECTED_RESULT_V1.json` with its outcome-access receipt, and the comparator-fair disjoint second execution in `v2/P1_X_PROTECTED_RESULT_V2.json` with `v2/P1_X_INDEPENDENT_VERIFICATION_V2.json`. The case schema is `research/claim_expansion/p1/SCIENTIFIC_REVISION_RESPONSIBILITY_CASE_V1.schema.json`.

12.6 Bounded meta-discovery studies

The study calculus and the three decomposition pilots are under `development/recursive-atom-calculus-closure-v1/` and `development/jump-atom-pilot-closure-v1/`. The scoped failure-knowledge study is `development/failure-epistemology-prospective-closure-v1/`, and the protected zero-error representation-transition study is `research/extensions/orion-jump-recursive-atoms/zero_error_jump/`.

12.7 Claim ledgers and nearest work

The mapping from each result-bearing statement in this manuscript to the archived artifact that supports it is `evidence/CLAIM_LEDGER.md`, with the immutable record of the earlier revision at `evidence/CLAIM_LEDGER_V1.md`. The full nearest-work mechanism matrix, from which Section 9 is drawn, is `evidence/NEAREST_WORK_MATRIX_V2.md` with `evidence/nearest_work_matrix_v2.json`. The bounded readiness attestation is `evidence/PEER_REVIEW_READY_BOUNDED_V2.md`.

12.8 Repository access and reproduction

The implementation repository holds the source code, protocol definitions, case suite, and repository-held evidence artifacts used by this revision. Persistent DOI assignment remains pending publication. Result-bearing artifacts are identified by their repository paths and content bindings rather than by an unverified archival claim.

The exact decompression, independent-rescoring, byte-comparison, and figure-regeneration commands are given in `REPRODUCE.md`. The command

```
make paper01-results
```

regenerates the publication tables from archived raw records only. It does not execute systems under test or create missing external evidence. On a checkout without an admissible archived study run the underlying module reports the result as undetermined; malformed or unbound archives are refused rather than converted into publishable numbers.

A fresh code and test environment requires Python 3.11 or later, an editable install with the development extras from the repository root, and model-provider access only for commands that actually execute a live-provider arm; dependencies are declared in `pyproject.toml`.

12.9 Runtime and cost

The credential-free mutation-necessity campaign and its independent verification each complete in approximately one minute on the current CPU environment. They make no network call and incur no model-provider cost. Algorithmic effort is fixed at 2,882 worlds, nine runnable arms, five ablations, and 10,000 registered paired-bootstrap resamples. The provider cost of the earlier live-provider arm was not recorded under a bound environment and therefore remains undetermined; it is not imputed from the mechanical successor.

12.10 Licensing status

This revision does not contain a repository-level licence file or a project licence declaration in `pyproject.toml`. Redistribution terms for code, case definitions, and protocol artifacts therefore remain undetermined from the repository state and must be resolved explicitly before archival or publication. Model-provider terms remain separately governed by the relevant providers.

12.11 Artifact fingerprinting

Result-bearing execution identities are bound through the protocol and archived records. Relevant coordinates include dataset and suite revisions and fingerprints; baseline configuration hashes; evaluator identity; frozen split hashes; and the exact subject revision and provider/model revisions for a live run. Reproduction of an empirical result requires the exact bound subject and admissible archived records; prose or a protocol declaration alone is not evidence that the run occurred.

13 Ethics, safety, and resources

13.1 Ethical considerations

This work studies mechanistic aspects of research workflow design rather than reporting a deployment in a human-facing domain. The frozen local evaluation uses constructed cases and does not require human-subject records or personally identifiable information. Any later external-provider or retrieval execution must be documented separately from this local evidence rather than inferred from the protocol design.

13.2 Safety considerations

Authority promotion and formulation reconstruction are separate concerns, and only the second is studied here. The mechanisms described can expose a possible method failure but cannot authorize M_t to rewrite itself. The reframe gate is mechanically bounded: only formulation and search-universe coordinates may trigger a reframe, and the evidence-only and execution-only negative controls exercise the corresponding non-reframe behaviour in the frozen local suite.

13.3 Resource usage and environmental impact

The repository does not yet contain an admissible clean-environment resource ledger for the full campaign. Numerical model-token, runtime, monetary-cost and carbon-impact quantities therefore remain undetermined in this revision. A future result-bearing run must retain actual resource usage under its frozen provider and environment bindings before those quantities are reported.

13.4 Dual-use considerations

The epistemic-state machinery described here is general-purpose and could be applied outside scientific research. This paper evaluates scientific-workflow mechanics and does not present an empirical evaluation of military or other high-risk deployment applications. The authority boundaries above are part of the implemented design, but they should not be read as evidence of safety in domains that were not evaluated.

14 Conclusion

This paper treats recursive research as an epistemic-state problem rather than as a synonym for repeated model calls. The mechanism separates the knowledge state, the relevance and search-universe model, and the governed method state; it diagnoses typed residuals so that responsibility constrains the authority to rewrite a formulation coordinate; and it reopens only the closure that depends on a changed coordinate. Recursive audit, active diagnosis, counterfactual minimal repair, diagnosis-to-action admission, certificate enforcement, dependency rollback, evidence-gated revision, repair assignment, model-class expansion and generic mutation authorization are absorbed donor substrate rather than contributions. What is put forward and tested is the composition: lower-level exclusion, a high-level counterfactual effect, protected-invariant preservation, and exact dependency-impact binding before a scientific $K/W/M$ mutation.

The powered mutation-necessity experiment supports that bounded composition on its credential-free mechanical family. Across a prospectively frozen primary run and a disjoint replication, the governed policy exceeds all three strong parents by more than 0.50 protected hidden-shift success, preserves perfect control restraint and protected success, and passes every frozen non-regression, safety, multiplicity and cost gate. Direct ablations reproduce the intended mechanism failures, and an independent implementation recomputes every score and analysis field without disagreement.

The revision-responsibility battery widens the evidence from one hidden-shift mechanism to the control problem of scientific revision responsibility. In a separately frozen, comparator-fair disjoint execution, the escalation contract achieves 400 of 400 exact revision decisions versus 275 of 400 for a donor-complete greedy product, a paired advantage of 0.3125 with 95% CI [0.2675,0.3575], while preserving zero false high-level reframes and zero protected-invariant violations. The effect is positive in every exact domain and survives the tested equally informative diagnosis and recovery ordering substitutions. An independent implementation reproduces the protected identities, result hashes and headline counts.

The strongest boundary is equally important: an ideal donor product supplied exactly the same scientific responsibility, minimality, preservation, reopen and authority semantics ties at 400 of 400. The contribution is therefore not inherent expressivity or ownership of the component mechanisms. It is the bounded evidence that an explicit minimal scientific-escalation contract can improve revision decisions over a realistic donor-complete interface that lacks that coupling. These results justify a claim over the registered mechanical and exact heterogeneous contract families; they do not justify model-general, open-ended, retrieval, or deployed autonomous-science superiority, which remain explicit external boundaries.

The bounded meta-discovery programme sharpens the same donor-first lesson. Its study calculus survives, but the first broad labels for tension, thought experiment and concept each split into several candidate mechanics rather than becoming universal new primitives. Scoped failure knowledge is useful while standard failure-learning parents remain sufficient for the tested incremental effect. Most decisively, on the protected zero-error representation-transition study the candidate composition and the strongest verified regime-revision parent tie exactly on both disjoint frozen splits, including zero false transitions. Those outcomes are recorded as decomposition and parent subsumption: they improve the boundary on when to investigate or reopen a formulation, but they do not add a new general $K/W/M$ mutation operator or a scientific-creativity claim.

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